Name: _____

TOPIC TEST 2

Chapter 1 Derivatives, exponential and trigonometric functions

- Time allowed: 70 minutes
- Part A: Five multiple-choice questions (5 marks).
- Part B: Six short-answer questions (51 marks).
- Part C: Two analysis questions (22 marks).

Part A

Multiple-choice questions

Five multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 Which statement is *not* true?

A $\frac{d}{dx}[\sin(5x)] = 5\cos(5x)$

B $\frac{d}{dx}(e^{2x}) = 2e^{2x}$

C $\frac{d}{dx}[\tan(3x)] = \frac{1}{3\cos^2(3x)}$

D $\frac{d}{dx}[\cos(7x)] = -7\sin(7x)$

E $\frac{d}{dx}[2x + 5]^2 = 4(2x + 5)$

[1 mark]

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2 The derivative of $(8x + 1)(3x - 4)^4$ is:

A $8(3x - 4)^4 + 12(8x + 1)(3x - 4)^3$

B $96(3x - 4)^3$

C $24(3x - 4)^3$

D $8(3x - 4)^4 + 4(8x + 1)(3x - 4)^3$

E $8(3x - 4)^4 + 3(8x + 1)(3x - 4)^3$

[1 mark]

3 What is the derivative of $\frac{x^2 + 1}{2x - 3}$?

A $\frac{2(x^2 - 3x + 1)}{(2x - 3)^2}$

B $\frac{2(3x^2 - 3x + 1)}{(2x - 3)^2}$

C $\frac{2(x^2 - 3x - 1)}{(2x - 3)^2}$

D $\frac{-2(x^2 - 3x - 1)}{(2x - 3)^2}$

E $\frac{-2(x^2 - 3x + 1)}{(2x - 3)^2}$

[1 mark]

4 A particle has displacement $x = 2 \cos(3t)$ cm over time t seconds. Find its acceleration after 5 seconds.

A 1.5 cm/s^2

B -1.5 cm/s^2

C -13.7 cm/s^2

D -11.7 cm/s^2

E 13.7 cm/s^2

[1 mark]

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5 The gradient of the tangent to the curve $y = \tan 2x$ at the point where $x = \frac{\pi}{6}$ is:

A $\frac{1}{3}$

B $\frac{2}{3}$

C 8

D 3

E $\frac{2}{\sqrt{3}}$

[1 mark]

Part B

Short-answer questions

Six short-answer questions
51 marks
Show your working where appropriate.

1 Differentiate:

a $(2x^3 - 7x + 3)^9$

b $\frac{2}{3x^5}$

c $\frac{1}{\sqrt{2x + 1}}$

d $\dfrac{x+1}{x-1}$

e $4x^2(2x-3)^2$

[13 marks]

2 Find the derivative of:

a e^{6x}

b $\cos(\pi x)$

c $\sin\left(\dfrac{x}{5}\right)$

d $(x + e^x)^4$

e $\dfrac{\tan(x)}{x}$

[8 marks]

3 Find the exact gradient of the tangent to each curve.

a $y = 3\sin(x)$ at $x = \dfrac{\pi}{4}$

b $y = \tan\left(\dfrac{x}{3}\right)$ at $x = \pi$

c $y = e^{7x}$ at $x = 1$

d $y = 5\cos(4x)$ at $x = \frac{\pi}{12}$

[9 marks]

4 Find the equation of the tangent to the curve $y = \cos\left(\frac{2x}{3}\right)$ at the point $\pi\left(\pi, -\frac{1}{2}\right)$.

[6 marks]

5 The gradient of the tangent to the curve $y = 2\sin \pi x$ at the point $P(x_1, y_1)$ is $\sqrt{3}\pi$. Find two possible coordinates for P.

[6 marks]

6 Plutonium decays according to the formula $M = M_0 e^{-0.008t}$ where M is the mass of Plutonium remaining in grams and t is time in years.

a If plutonium has a mass of 100 g, what will its mass be after:

i 10 years?

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ii 80 years?

iii 150 years?

b At what rate is the plutonium decaying after:

i 10 years?

ii 80 years?

iii 150 years?

c If the half-life of a decaying substance is the time taken to halve the mass of unchanged substance, find the approximate half-life of plutonium.

[9 marks]

Part C

Analysis questions

Two analysis questions
22 marks
Show your working where appropriate.

1 A pendulum moving in simple harmonic motion has an equation given by $x = 4 \sin\left(\frac{\pi t}{2}\right)$.

a When is the pendulum in equilibrium?

b What is the maximum displacement from the point of equilibrium?

c Find the velocity of the pendulum when it is:

i in equilibrium

ii at the maximum displacement.

d Find the acceleration of the pendulum when it is:

i in equilibrium

ii at the maximum displacement.

e Describe the movement of the pendulum, relating its velocity and acceleration to different displacements.

[17 marks]

2 A spherical metal ball is heated so that its radius is expanding at the rate of 0.04 mm/second.

a From the formula for the volume of a sphere, find the rate of change in volume in terms of its radius r.

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b The chain rule can be written as $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$. Use this rule to find the rate at which the volume changes over time t when its radius is 3.4 mm.

[5 marks]

[Total marks: 78]

Answers

Part A – Multiple-choice questions [1 mark each]

1 $\frac{d}{dx}[\tan(3x)] = 3 \times \frac{1}{\cos^2(3x)} = \frac{3}{\cos^2(3x)}$ so statement C is not true.

$\therefore$ **C**

2 $\frac{d}{dx}[(8x+1)(3x-4)^4] = (3x-4)^4 \times 8 + (8x+1) \times 3 \times 4(3x-4)^3$

$= 8(3x-4)^4 + 12(8x+1)(3x-4)^3$

$\therefore$ **A**

3 $\frac{d}{dx}\left(\frac{x^2+1}{2x-3}\right) = \frac{(2x-3) \times 2x - (x^2+1) \times 2}{(2x-3)^2}$

$= \frac{4x^2 - 6x - 2x^2 - 2}{(2x-3)^2}$

$= \frac{2x^2 - 6x - 2}{(2x-3)^2}$

$= \frac{2(x^2 - 3x - 1)}{(2x-3)^2}$

$\therefore$ **C**

4 $x = 2\cos(3t)$

$v = -6\sin(3t)$

$a = -18\cos(3t)$

When $t = 5$:

$a = -18\cos(15)$

$= 13.7$

$\therefore$ **E**

5 $\frac{d}{dx}[\tan(2x)] = \frac{2}{\cos^2(2x)}$

When $x = \frac{\pi}{6}$:

$\frac{dy}{dx} = \frac{2}{\cos^2\left(2 \times \frac{\pi}{6}\right)}$

$= \frac{2}{\cos^2\left(\frac{\pi}{3}\right)}$

$= \frac{2}{\left(\frac{1}{2}\right)^2}$

$$= \frac{2}{\frac{1}{4}}$$
$$= 8$$

$\therefore$ C

Part B – Short-answer questions

1 a $\frac{d}{dx}[(2x^3 - 7x + 3)^9] = (6x^2 - 7) \times 9(2x^3 - 7x + 3)^8$ **[1 mark]**

$= 9(6x^2 - 7)(2x^3 - 7x + 3)^8$ **[1 mark]**

b $\frac{d}{dx}\left(\frac{2}{3x^5}\right) = \frac{d}{dx}\left(\frac{2}{3}x^{-5}\right)$ **[1 mark]**

$= -\frac{10}{3}x^{-6}$ **[1 mark]**

$= -\frac{10}{3x^6}$ **[1 mark]**

c $\frac{d}{dx}\left(\frac{1}{\sqrt{2x+1}}\right) = \frac{d}{dx}(2x+1)^{-\frac{1}{2}}$ **[1 mark]**

$= 2 \times -\frac{1}{2}(2x+1)^{-\frac{3}{2}}$ **[1 mark]**

$= -\frac{1}{(2x+1)^{\frac{3}{2}}}$

$= -\frac{1}{\sqrt{(2x+1)^3}}$ **[1 mark]**

d $\frac{d}{dx}\left(\frac{x+1}{x-1}\right) = \frac{(x-1) \times 1 - (x+1) \times 1}{(x-1)^2}$ **[1 mark]**

$= \frac{x - 1 - x - 1}{(x-1)^2}$

$= \frac{-2}{(x-1)^2}$ **[1 mark]**

e $\frac{d}{dx}[4x^2(2x-3)^2] = (2x-3)^2 \times 8x + 4x^2 \times 2 \times 2(2x-3)^1$ **[1 mark]**

$= 8x(2x-3)^2 + 16x^2(2x-3)$ **[1 mark]**

$= 8x(2x-3)[2x - 3 + 2x]$

$= 8x(2x-3)(4x-3)$ **[1 mark]**

2 a $\frac{d}{dx}(e^{6x}) = 6e^{6x}$ **[1 mark]**

b $\frac{d}{dx}[\cos(\pi x)] = -\pi \sin(\pi x)$ **[1 mark]**

c $\frac{d}{dx}\left[\sin\left(\frac{x}{5}\right)\right] = \frac{1}{5}\cos\left(\frac{x}{5}\right)$ **[1 mark]**

d $\frac{d}{dx}[(x + e^x)^4] = (1 + e^x) \times 4(x + e^x)^3$ **[1 mark]**

$= 4(1 + e^x)(x + e^x)^3$ **[1 mark]**

e $\frac{d}{dx}\left[\frac{\tan(x)}{x}\right] = \frac{x \times \frac{1}{\cos^2(x)} - \tan(x) \times 1}{x^2}$ **[1 mark]**

$$= \frac{\frac{x}{\cos^2(x)} - \frac{\tan(x)\cos^2(x)}{\cos^2(x)}}{x^2}$$

$$= \frac{\frac{x - \tan(x)\cos^2(x)}{\cos^2(x)}}{x^2}$$

$$= \frac{x - \tan(x)\cos^2(x)}{x^2\cos^2(x)}$$ **[2 marks]**

3 a $\frac{d}{dx}[3\sin(x)] = 3\cos(x)$ **[1 mark]**

When $x = \frac{\pi}{4}$, $\frac{dy}{dx} = 3\cos\left(\frac{\pi}{4}\right)$

$= 3 \times \frac{1}{\sqrt{2}}$

$= \frac{3}{\sqrt{2}}$ **[1 mark]**

b $\frac{d}{dx}\left[\tan\left(\frac{x}{3}\right)\right] = \frac{1}{3} \times \frac{1}{\cos^2\left(\frac{x}{3}\right)}$

$= \frac{1}{3\cos^2\left(\frac{x}{3}\right)}$ **[1 mark]**

When $x = \pi$, $\frac{dy}{dx} = \frac{1}{3\cos^2\left(\frac{\pi}{3}\right)}$

$= \frac{1}{3 \times \left(\frac{1}{2}\right)^2}$

$= \frac{1}{3 \times \frac{1}{4}}$

$= \frac{4}{3}$ **[1 mark]**

c $\frac{d}{dx}(e^{7x}) = 7e^{7x}$ **[1 mark]**

When $x = 1, \frac{dy}{dx} = 7e^{7}$ **[1 mark]**

d $\frac{d}{dx}[5\cos(4x)] = -20\sin(4x)$ **[1 mark]**

When $x = \frac{\pi}{12}, \frac{dy}{dx} = -20\sin\left(4 \times \frac{\pi}{12}\right)$

$= -20\sin\left(\frac{\pi}{3}\right)$

$= -20 \times \frac{\sqrt{3}}{2}$

$= -10\sqrt{3}$ **[2 marks]**

4 $\frac{d}{dx}\left[\cos\left(\frac{2x}{3}\right)\right] = -\frac{2}{3}\sin\left(\frac{2x}{3}\right)$ **[1 mark]**

At $\left(\pi, -\frac{1}{2}\right), \frac{dy}{dx} = -\frac{2}{3}\sin\left(\frac{2\pi}{3}\right)$

$= -\frac{2}{3}\sin\left(\pi - \frac{\pi}{3}\right)$ (2nd quadrant)

$= -\frac{2}{3}\sin\left(\frac{\pi}{3}\right)$

$= -\frac{2}{3} \times \frac{\sqrt{3}}{2}$

$= -\frac{\sqrt{3}}{3}$ **[2 marks]**

$y - y_1 = m(x - x_1)$

$y - \left(-\frac{1}{2}\right) = -\frac{\sqrt{3}}{3}(x - \pi)$ **[1 mark]**

$6y + 3 = -2\sqrt{3}(x - \pi)$

$= -2\sqrt{3}x + 2\sqrt{3}\pi$

$2\sqrt{3}x + 6y + 3 - 2\sqrt{3}\pi = 0$ **[2 marks]**

5 $\frac{d}{dx}[2\sin(\pi x)] = 2\pi\cos(\pi x)$ **[1 mark]**

$2\pi\cos(\pi x) = \sqrt{3}\pi$ **[1 mark]**

$\cos(\pi x) = \frac{\sqrt{3}\pi}{2\pi} = \frac{\sqrt{3}}{2}$

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$$\pi x = \frac{\pi}{6}, 2\pi - \frac{\pi}{6}, 2\pi + \frac{\pi}{6}, 2\pi + \left(2\pi - \frac{\pi}{6}\right), \ldots \quad \text{(1st and 4th quadrants)}$$

$$= \frac{\pi}{6}, \frac{11\pi}{6}, \frac{13\pi}{6}, \frac{23\pi}{6}, \ldots$$

$$x = \frac{1}{6}, \frac{11}{6}, \frac{13}{6}, \frac{23}{6}, \ldots$$ **[2 marks]**

When $x = \frac{1}{6}$, $y = 2 \sin\left(\frac{\pi}{6}\right)$

$$= 2 \times \frac{1}{2}$$

$$= 1$$ **[1 mark]**

When $x = \frac{11}{6}$, $y = 2 \sin\left(\frac{11\pi}{6}\right)$

$$= 2 \times -\frac{1}{2} \text{ (4th quadrant)}$$

$$= -1$$ **[1 mark]**

Two possible coordinates are $\left(\frac{1}{6}, 1\right), \left(\frac{11}{6}, -1\right)$

6 Substitute $M = 100$, $t = 0$:

$100 = M_0 e_0$

$= M_0$

So $M = 100e^{-0.008t}$ **[1 mark]**

a i Substitute $t = 10$

$M = 100e^{-0.008 \times 10}$

$= 92.3$ g **[1 mark]**

ii Substitute $t = 80$

$M = 100e^{-0.008 \times 80}$

$= 52.7$ g **[1 mark]**

iii Substitute $t = 150$

$M = 100e^{-0.008 \times 150}$

$= 30.1$ g **[1 mark]**

b $\frac{d}{dx}(100e^{-0.008t}) = 100 \times -0.008e^{-0.008t}$

$= -0.8e^{-0.008t}$ **[1 mark]**

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i Substitute $t = 10$

$$\frac{dy}{dx} = -0.8e^{-0.008 \times 10}$$

$$= -0.74 \text{ g/year}$$ **[1 mark]**

ii Substitute $t = 80$

$$\frac{dy}{dx} = -0.8e^{-0.008 \times 80}$$

$$= -0.42 \text{ g/year}$$ **[1 mark]**

iii Substitute $t = 150$

$$\frac{dy}{dx} = -0.8e^{-0.008 \times 150}$$

$$= -0.24 \text{ g/year}$$ **[1 mark]**

c A 100 g mass decays to 52.7 g after 80 years, so the half-life is just over 80 years.

Checking some values:

$t = 85$: $M = 100e^{-0.008 \times 85} = 50.66$ g

$t = 86$: $M = 100e^{-0.008 \times 86} = 50.26$ g

$t = 87$: $M = 100e^{-0.008 \times 87} = 49.86$ g

So the half-life is about 87 years. **[1 mark]**

Part C – Analysis questions

1 a Substitute $x = 0$:

$$4 \sin\left(\frac{\pi t}{2}\right) = 0$$ **[1 mark]**

$$\sin\left(\frac{\pi t}{2}\right) = 0$$

$$\frac{\pi t}{2} = 0, \pi, 2\pi, 3\pi, \ldots$$

$t = 0, 2, 4, 6, \ldots$ s **[2 marks]**

b Amplitude = 4 so the maximum displacement is 4 (at $x = \pm 4$) **[1 mark]**

c $v = 4 \times \frac{\pi}{2} \cos\left(\frac{\pi t}{2}\right)$

$$= 2\pi \cos\left(\frac{\pi t}{2}\right)$$ **[1 mark]**

i When $x = 0$, $t = 0, 2, 4, 6, \ldots$

Substitute $t = 0$: $v = 2\pi \cos(0) = 2\pi$

Substitute $t = 2$: $v = 2\pi \cos(\pi) = -2\pi$

Substitute $t = 4$: $v = 2\pi \cos(2\pi) = 2\pi \ldots$

The velocity when the pendulum is in equilibrium is $\pm 2\pi$ cm/s **[2 marks]**

ii When $x = \pm 4$:

$4 \sin\left(\frac{\pi t}{2}\right) = \pm 4$

$\sin\left(\frac{\pi t}{2}\right) = \pm 1$

$\frac{\pi t}{2} = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \ldots$

$t = 1, 3, 5, \ldots$ s **[2 marks]**

Substitute $t = 1$: $v = 2\pi \cos\left(\frac{\pi}{2}\right) = 0$

Substitute $t = 3$: $v = 2\pi \cos\left(\frac{3\pi}{2}\right) = 0$

Substitute $t = 5$: $v = 2\pi \cos\left(\frac{5\pi}{2}\right) = 0 \ldots$

The velocity when the pendulum is at its maximum displacement is 0 cm/s. **[2 marks]**

Alternatively: At maximum displacement, the pendulum reverses direction, so the velocity must be 0.

d $a = 2\pi \times -\frac{\pi}{2} \sin\left(\frac{\pi t}{2}\right)$

$= -\pi^2 \sin\left(\frac{\pi t}{2}\right)$ **[1 mark]**

i When $x = 0$, $t = 0, 2, 4, 6, \ldots$

Substitute $t = 0$: $a = -\pi^2 \sin 0 = 0$

Substitute $t = 2$: $a = -\pi^2 \sin \pi = 0$

Substitute $t = 4$: $a = -\pi^2 \sin 2\pi = 0 \ldots$

The acceleration when the pendulum is in equilibrium is 0 cm/s^2. **[2 marks]**

ii When $x = \pm 4$, $t = 1, 3, 5, 7, \ldots$

Substitute $t = 1$: $a = -\pi^2 \sin\left(\frac{\pi}{2}\right) = -\pi^2$

Substitute $t = 3$: $a = -\pi^2 \sin\left(\frac{3\pi}{2}\right) = \pi^2$

Substitute $t = 5$: $a = -\pi^2 \sin\left(\frac{5\pi}{2}\right) = -\pi^2 \ldots$

The acceleration when the pendulum is at its maximum displacement is $\pm\ \pi^2$ cm/s^2. **[2 marks]**

e The pendulum is in equilibrium at 0, 2, 4, 6, … seconds. While in equilibrium, its velocity is $\pm 2\pi$ and acceleration is 0. At its maximum displacement of ± 4, the velocity is 0 and the acceleration is $\pm\pi^2$. As the pendulum moves through equilibrium it has its greatest velocity and at the maximum displacement it is stationary at that instant, then moves back the other way. The acceleration is zero when it is in the centre and its maximum out at the edge of the movement. **[1 mark]**

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2 a $V = \frac{4}{3}\pi r^3$ **[1 mark]**

$\frac{dV}{dr} = 4\pi r^2$ **[1 mark]**

b $\frac{dr}{dt} = 0.04$ **[1 mark]**

$$\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$$
$$= 4\pi r^2 \times 0.04$$
$$= 0.16\pi r^2$$ **[1 mark]**

Substitute $r = 3.4$:

$$\frac{dV}{dt} = 0.16\pi \times 3.4^2$$
$$= 5.81$$ **[1 mark]**

The volume changes at the rate of 5.81 mm^3/s.

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